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Supplementary information

Real-time multi-angle projection imaging of biological dynamics

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Supplementary Information

Real-Time Multi-Angle Projection Imaging of Biological Dynamics

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Supplementary Notes

Supplementary Note 1: PSF Engineering Approaches

Various methods have been developed to extend the depth of focus of an imaging system. In the realm of fluorescence imaging, phase masks are prefered over amplitude masks, such that the fluorescence light over the entire pupil can be used[1, 2]. The modification of the wavefront of the incoming beam lowers the Strehl ratio of the imaging system and in all practical implementations, also reduces the spatial resolution. One way to imagine this relationship is as follows: in order to extend the PSF n-fold, the energy contained in the diffraction limited PSF has to be stretched out n-fold as well. As such, the peak intensity (which is one way the Strehl ratio is defined) one can obtain from a single emitter will become smaller with PSF engineering.

Extended Data Figure 1 shows simulated extended-depth-of-field (EDF) acquisitions alongside simulated acquisitions for our projection method as applied to OPM and wide-field microscopy. These were created by convolving mitochondrial membrane segmentations of the jrc_hela-2 dataset from openorganelle.org[3] with appropriate point spread functions, simulated using Debye vectorial diffraction code[4]. For this dataset, the mitochondrial network spans approximately 5.4 microns in depth, and so we engineered EDFs with a range of ~6 microns. For EDF methods, the image shown is a slice taken from the centre of the fully convolved 3D volume, corresponding to the image on the camera recorded without moving the sample. For Pro methods, the image shown is the sum of all slices, acquired during one camera exposure, while the sample is moved in *z*.

For all simulations, the excitation and emissions wavelengths were 488 nm and 510 nm, respectively, while the voxel size was 64 nm × 64 nm × 64 nm. The 's3-level' mitochondrial membrane segmentation from openorganelle.org was bicubically downsampled from 32 nm × 41.92 nm × 32 nm to match this and padded with empty planes in *z* to avoid artefacts from the circulant nature of the convolution employed. Except for the 'low NA WF' simulation (for which a numerical aperture of 0.5 was used), all simulations assumed an objective numerical aperture of 1.35 and a refractive index of 1.4. No noise was added following convolution, to show all methods in their best possible, high signal-to-noise ratio, case.

Of all the methods considered, OPM-Pro (shown in **Extended Data Figure 1B**) provides the clearest image. The image and associated effective point spread function (PSF) are also the most amenable to deconvolution, with 200 iterations of the unregularised Richardson-Lucy algorithm producing a result closer to the ground truth image than any other method. WF-Pro (**Extended Data Figure 1C**) provides a clearer image than two 'traditional' EDF techniques, the layer cake of Abrahamsson et al. (**Extended Data Figure 1e**)[2] and the cubic phase mask of Dowski and Cathey (**Extended Data Figure 1F**)[5]. In both EDF cases, this is due to the significant contribution of sidelobes in the PSF. The layer cake PSF is effectively the sum of 12 Bessel PSFs (as the depth-of-field extension scales with the number of layers and hence Bessel PSFs, as proven in Chen et al.[6]), while the cubic phase mask introduces the sidelobes characteristic of Airy beams. While the out-of-focus haze produced in the layer cake case can be effectively removed by a 2D deconvolution, this is not possible in the cubic phase case due to the depth-varying nature of the sidelobes.

Simply stopping down the numerical aperture of the lens used to image the sample, in this case from 1.35 to 0.5, extends the depth-of-field of the system to the requisite 5.5 microns, without introducing any sidelobes, providing the haze-free result shown in **Extended Data Figure 1D.** While the deconvolved result initially looks sharp, the dual-layer structure of the mitochondrial membrane has been collapsed into a single line, a typical artefact of unregularised iterative deconvolution methods such as Richardson-Lucy. In addition,

stopping down the numerical aperture so severely would lead to a significant reduction in the amount of light collected (theoretically greater than 90 %).

These visual observations are supported by the spectral analysis presented in **Extended Data Figure 2**. This presents logarithmically-scaled Fourier power spectra (a-e), along with a radially-averaged profiles (f). Here we see that OPM-Pro provides the best transfer out to the diffraction limit, with the cubic phase mask providing the worst transfer at all spatial frequencies. While it may appear that the low NA wide-field case is superior to the other methods at low spatial frequencies (i.e. at small fractions of the diffraction limit), it is important to note that these profiles are all normalised to start at 1. In reality, the much reduced collection efficiency associated with this method means that it would, in practice, start closer to 0.1 and decrease in a similar fashion.

Supplementary Note 2: Comparison to Physical Sample Rotation and mutliple viewing directions

Physical sample rotation can enable isotropic resolution even if the underlying microscope system has highly anisotropic 3D resolution. Further, by fusing data from multiple views, the deleterious effects of shadowing, optical occlusions and to some degree optical scattering can be reduced. The overall 3D resoluiton, if sufficient angular sampling is applied, approaches the lateral resolution of the parent microscope technology. In the context of light-sheet microscopy, depending on the numerical aperture applied to light-sheet generation and detection, typically 6-8 rotation angles are needed to isotropically fill reciprocal space with sample information[7]. The obvious drawback is that sample rotation is slow and can only be implemented when sufficient working distance for appropriate sample mounting is present. An alternative approach exists with microscope systems that provide multiple physical viewing directions. In the field light-sheet fluorescence microscopy, techniques such as diSPIM[8] or isoView[9] typically provide two orthogonal detection axes (while isoView employs four objectives, the available axes are two). While this enables faster data acquisiton than physical sample rotation, the limitation to two detection axes lead to an incomplete coverage of reciprocal space, and technically non-uniform resolution along different spatial axes. Further, for such systems the numerical aperture is curtailed to 0.7 to 0.8 due to steric hindrance, which reduces sensitivity and resolution. Lastly, such systems are not compatible with standard sample mounting and are much more complex to integrate in an exisitng microscope.

Our projection method instead maintains the 3D resolution of its parent microscope technology and produces projections from it without requiring sample rotation or additional viewing directions. Being reliant on the 3D resolving power of the parent technology, our method will in most cases inherit the anisotropic 3D resolution of a typical microsocpe. Notably, some light-sheet microscopes achieve isotropic[10, 11] or near-isotropic[12, 13] 3D resolution through advanced light-sheet engineering. Thus they represent a path to obtaining isotropic or close to isotropic views with our projection method.

Supplementary Note 3: Equivalence between the projection from a rotated and a sheared volume.

For the application in LLSM and OPM, our scan unit can perform the necessary shearing to exactly undo the lateral displacement that occurs during the scanning process in those two microscopes. As such, the resulting projection is expected to be equivalent to the numerically sheared and projected 3D data coming from such a microscope. But what happens if the scan unit performs a larger or smaller shearing angle than what is necessitated by the microscope? This general case, i.e. shearing by an arbitrary angle and subsequent projection, is investigated below. Consider, without loss of generality, shearing along the x-axis by a and rotating in a positive sense around the z-axis by Θ. The rotation matrix is given by

$$R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

and the shear matrix by

$$S = \begin{bmatrix} 1 & \alpha \\ 0 & 1 \end{bmatrix}$$

In both cases, we wish to project along the y-axis, for which the projection matrix is

$$P = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

Consider now the projection of the rotation matrix,

$$PR = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ 0 & 0 \end{bmatrix}$$

And the projection shear matrix,

$$PS = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & \alpha \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & \alpha \\ 0 & 0 \end{bmatrix}$$

If we multiply PS by $\cos\theta$, we can achieve equality if $\alpha = -\sin\theta / \cos\theta = -\tan\theta$.

The multiplication of $\cos\theta$, has the effect of scaling the output:

$$MPS = \begin{bmatrix} Sx & 0 \\ 0 & Sy \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & \alpha \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} Sx & \alpha Sx \\ 0 & 0 \end{bmatrix}$$

With $\alpha = -\tan\theta$ and Sx=$\cos\theta$, and M is the scaling matrix. However, in practice, the scan unit only performs the shearing, but not the scaling operation, i.e.

$$PS = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & \alpha \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & \alpha \\ 0 & 0 \end{bmatrix}$$

This has the following implications: Applied to lattice and OPM data, if we just undo the amount of shearing imparted by the diagonal scan trajectory, the projection is equivalent to a computationally sheared and projected dataset. However, applied to microscopes that do not need "de-skewing, or if we go away from the ideal shear parameter, the resulting projection image will be stretched in one dimension by $1/\cos\theta$, compared to a projection obtained from a volume that is numerically rotated. As such, the scaling operation (i.e., a squeezing of the image in one dimension) has to be done computationally, to make the projected views equivalent. For viewing purposes and for small angles (i.e. approximating $\cos\theta \sim 1$), however, this might be ignored.

Supplementary Note 4: Computational overhead for shearing and projecting 3D image stacks.

Acquiring a conventional z-stack and computationally processing it is a potential alternative option. However, performing such an operation in real-time is unlikely. Many of the 16-bit datasets presented in this manuscript, if imaged slice by slice, encompass 2048 x 2048 x 300 voxels (in X, Y, and Z, respectively), or approximately ~3GB. To acquire this volume at 100 Hz, necessitates the ability to save data at 300 GB/s, which is well beyond the capacity of (non-RAID) PCIe-based SSD devices (usually ~4GB/s) and even DDR4 SDRAM (~25.6GB/s). Thus, even from a data handling perspective, such processing is unlikely prior to affine shearing and sum projecting the data. Nonetheless, we provided some approximate benchmarks for the time necessary to perform such processing using a variety of CPU and GPU-based code. Our optimised CPU-bound C++ code took 250 ms to deskew a 2048 x 2048 x 300 voxel volume, operating on an i9-7900X, while a GPU-accelerated routine, using the Clij system for ImageJ, took 1670 ms to push the data to the GPU, 125 ms to apply the affine transform, and another 550 ms to pull the data back from the GPU, operating on an nVidia Tesla P40.

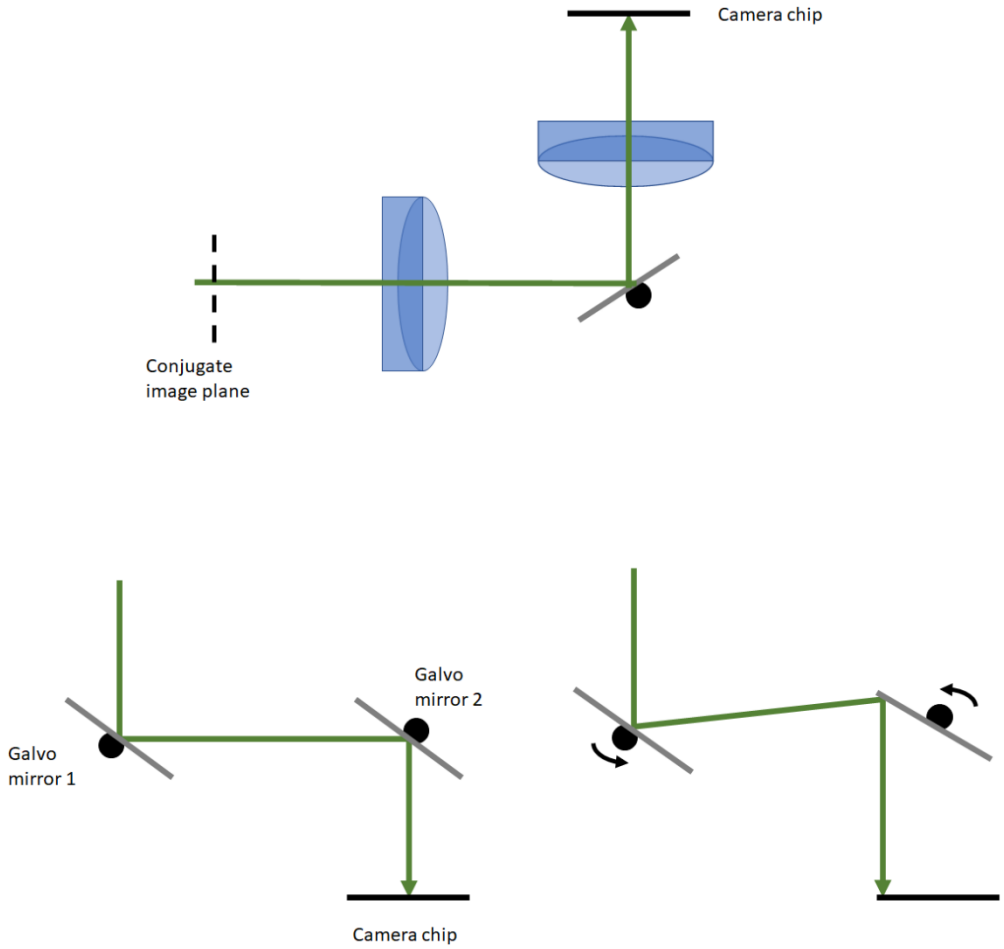
Supplementary Note 5: Design criteria for scan unit

The purpose of the scan unit is to rapidly translate the image on the camera detector. Ideally, this translation should be linear and telecentric (I.e. the translation is free of any rotation of the imaging rays impinging on the camera). There are in principle two ways to realize this shift rapidly: I) a relay telescope system consisting of two lense and one galvanometric or other scan mirror in its Fourier plane. II) Lensless scanning using two reflections from rotating mirrors.

I) is the traditional way to perform lateral scanning, but it comes with the caveat that the number of optical elements, in particular lenses, are increased in the system. This brings the risk of deteriorating the optical performance and losing light due to absorption and reflection losses. The method is however natural if the microscope system needs anyway a relay telescope for the camera (e.g. to increase the magnification to match the pixel size, e.g. to accommodate the larger pixel size of EMCCCD cameras).

II) Lensless scanning can reduce reflection and light absorption due to the absence of lenses while still maintaining telecentricity. This can be implemented with two mirrors, as in the work presented here, where the first mirror imparts an angular change in the beam path which is undone by the second mirror, resulting in a translation without angular change (see also **Supplementary Figure SN5.1**). It has been recently discovered that this scan and de-scanning can also be performed by a single galvanometric mirror, and a pair of static mirror that fold the beam path such that two reflections are made on the same galvanometric mirror[14]. This method can reduce the cost as only one galvanometric mirror is needed, however a larger mirror size is needed.

Due to better availability of smaller galvanometric mirrors with high scan speeds, we opted for the two scanning mirror solutions when implementing lensless scanning. Furthermore, we found that this implementation is relatively compact and simple to implement.

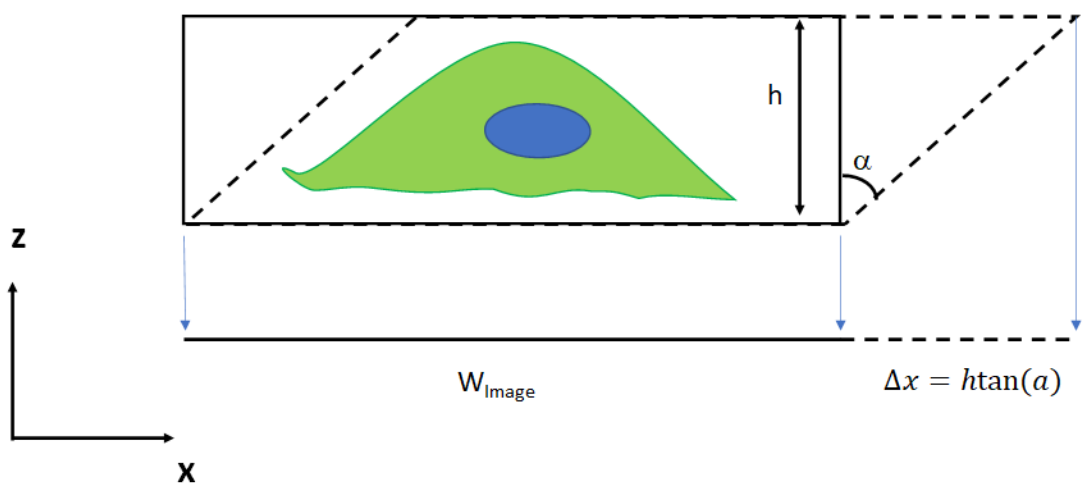


Supplementary Figure SN5.1 Methods for rapid shearing. The shearing unit needs to be able to rapidly translate the incoming light on the scale of the camera chip. Top shows a variant using a telescope and a galvanometric mirror located in the Fourier plane. Bottom shows lensless scanning using two mirrors that move in synchrony. The size of the second mirror needs to be the camera chip size times sqrt(2) in order to be able to cover the full camera chip.

Supplementary Note 6: Field of view considerations when using projected views.

When shearing is applied in the image formation, the resulting projection image becomes larger in one dimension compared to an unsheared image. The widening is proportional to the height h over which a projection is formed and the tangent of the shear angle (see also **Supplementary Figure SN6.1**). For the spinning disk data, the height was 10 microns and the available Field of view of our camera spanned 116 microns. To shear the image by plus and minus 45 degree, the shearing occupies 20 microns, or 17% of the available FOV. Another example is the Zebrafish imaging on our SPIM microscope. The heigh h was 100 microns, and the available FOV was 332microns. Shearing the image by 45 degrees reduces the available FOV by 100microns or ~30%.

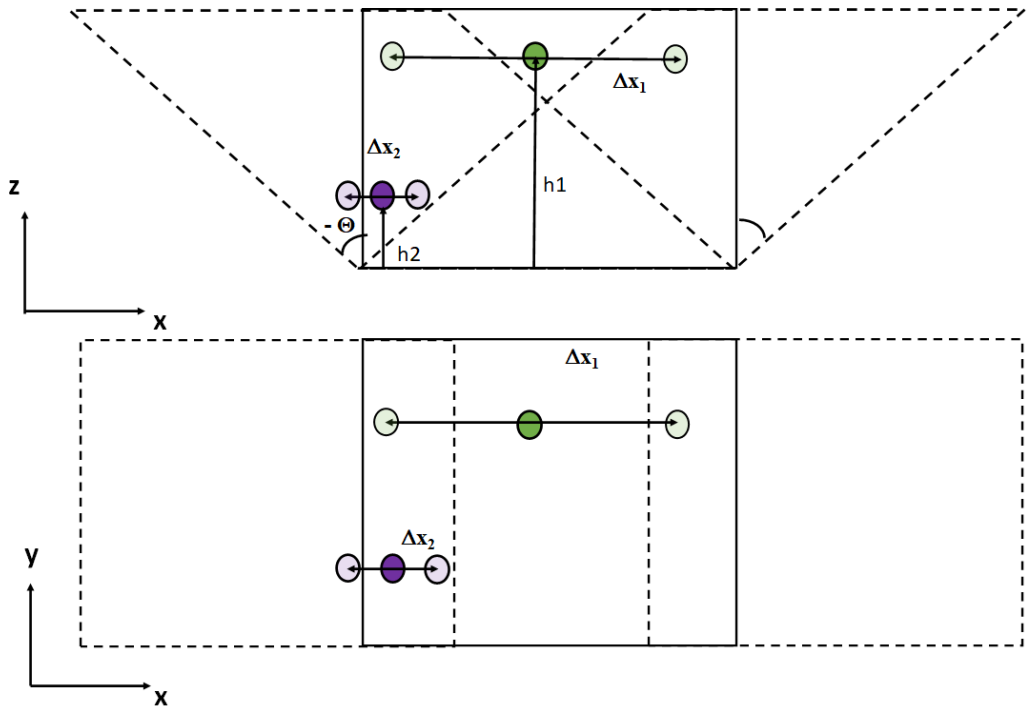
It has to be noted that newer sCMOS cameras have non square chip sizes. Thus the longer dimension could be used in the shearing direction. Our lensless shearing unit can access these larger chip sizes without additional aberrations, even if the optical field of view of the parent microscope does not cover this size.



Supplementary Figure SN6.1 Widening of the projection image upon shearing. The projection image is widened by Δx upon shearing, which is proportional to the height h of the volume over which a projection is taken and the tangent of the shear angle alpha.

Supplementary Note 7: Precision of particle localization using projected views.

Obtaining two views under different viewing angles allows triangulation of the 3D location of objects, provided the same object can be localized in each view. If this is the case, a simple reconstruction algorithm is schematically shown in **Supplementary Figure SN7.1**. The projection of two sheared volumes are taken, which have a shear angle of θ and - θ relative to the true, unsheared volume. One can find that the height of particles in this volume is given by: $h = \Delta x/2 \tan(\theta)$. Δx is the apparent displacement of an object between the two projections.



Supplementary Figure SN7.1 Determination of the z-position from two projections. Top: Two volumes sheared by $\pm\theta$ (dotted lines) and an unsheared volume (solid lines) are shown. In these three volumes, two particles, green and magenta, located at different heights are shown. Solid color depicts the particle's true position in the unsheared volume, and transparent colors depicts the particle's position in the sheared volumes. **Bottom:** Projection of the two volumes. The particle's displacement in the x direction, Dx, is proportional to their height.

We wish to calculate the theoretical 3D localization precision for a particle, given two views acquired using our projection method. We assume that the PSF is well-approximated by an ellipsoidal Gaussian function and that there is no background. Under these assumptions, the Thompson–Mortensen localization variance in each dimension is given by

$$\sigma^2 = \frac{16}{9}\frac{s^2 + a^2/12}{N},$$

where s is the standard deviation of the Gaussian function approximating the PSF, a is the pixel width and N is the number of photons collected[15]. The localization precision, σ, is given by the positive square root of this quantity.

For the field synthesis light sheet microscope used in this work, PSF measurements on sub-diffraction-sized beads provide full-widths at half-maximum of approximately 290x290x750nm in x, in y and in z, corresponding to s_x = 123 nm, s_y = 123 nm and s_z = 318nm. All acquisitions have a pixel size of 104 nm. In a projection acquisition, the PSF is rotated by an angle, θ, around the x-axis and also stretched in the y direction by a factor of $1/\cos\theta$. Hence, in the projection view (Pro shorthand for our projection method):

$$\begin{aligned} s_{x'}^{\mathrm{Pro}} &= s_x, \\ s_{y'}^{\mathrm{Pro}} &= \frac{1}{\cos\theta}\sqrt{\frac{s_z^2 s_y^2}{\left(s_z^2\cos^2\theta + s_y^2\sin^2\theta\right)}}, \end{aligned}$$

where x' and y' refer to the horizontal and vertical position in the camera frame, not 3D space.

Given two estimators, $\hat{\alpha}_1$ and $\hat{\alpha}_2$, of α, with variances σ_1^2 and σ_2^2, then, for the combined estimator

$$\hat{\alpha} = \frac{\sigma_2^2\,\hat{\alpha}_1 + \sigma_1^2\,\hat{\alpha}_2}{\sigma_1^2 + \sigma_2^2},$$

the variance of $\hat{\alpha}$ is given by

$$\frac{\sigma_1^2\sigma_2^2}{\sigma_1^2 + \sigma_2^2}.$$

For simplicity, we consider our projection acquisition scheme where both views are rotated by $\pm\theta$ from the conventional deskewed view. We further assume that both views correspond to the same number of integrated photons. Hence, per dimension, the localization precision in each view is equal, e.g. $\sigma_{x'_1}^{\mathrm{Pro}} = \sigma_{x'_2}^{\mathrm{Pro}}$.

As the apparent x co-ordinate is not changed by our projection acquisition procedure, the true x value can be estimated by pooling both x' measurements. Similarly, the true y value is estimated by considering the mean of the two y' measurements. The z location of the particle is given by

$$z = \frac{y'_2 - y'_1}{2\tan\theta}.$$

Hence,

$$\begin{aligned} \sigma_x^2 &= \frac{\left(\sigma_{y'}^{\mathrm{Pro}}\right)^2}{2} \\ \sigma_y^2 &= \frac{\left(\sigma_{y'}^{\mathrm{Pro}}\right)^2}{2} \\ \sigma_z^2 &= \frac{\left(\sigma_{y'}^{\mathrm{Pro}}\right)^2}{2\tan\theta}. \end{aligned}$$

We can compare this with a hypothetical situation in which two views at an angle $\pm\theta$ from the conventional deskewed view are acquired through a proper rotation of the volume, rather than a projection acquisition. In this situation, the stretching by $1/\cos\theta$ does not occur, giving

$$
\begin{aligned}
s_{x'}^{\mathrm{rot}} &= s_x, \\
s_{y'}^{\mathrm{rot}} &= \sqrt{\frac{s_z^2 s_y^2}{\left(s_z^2 \cos^2\theta + s_y^2 \sin^2\theta\right)}}.
\end{aligned}
$$

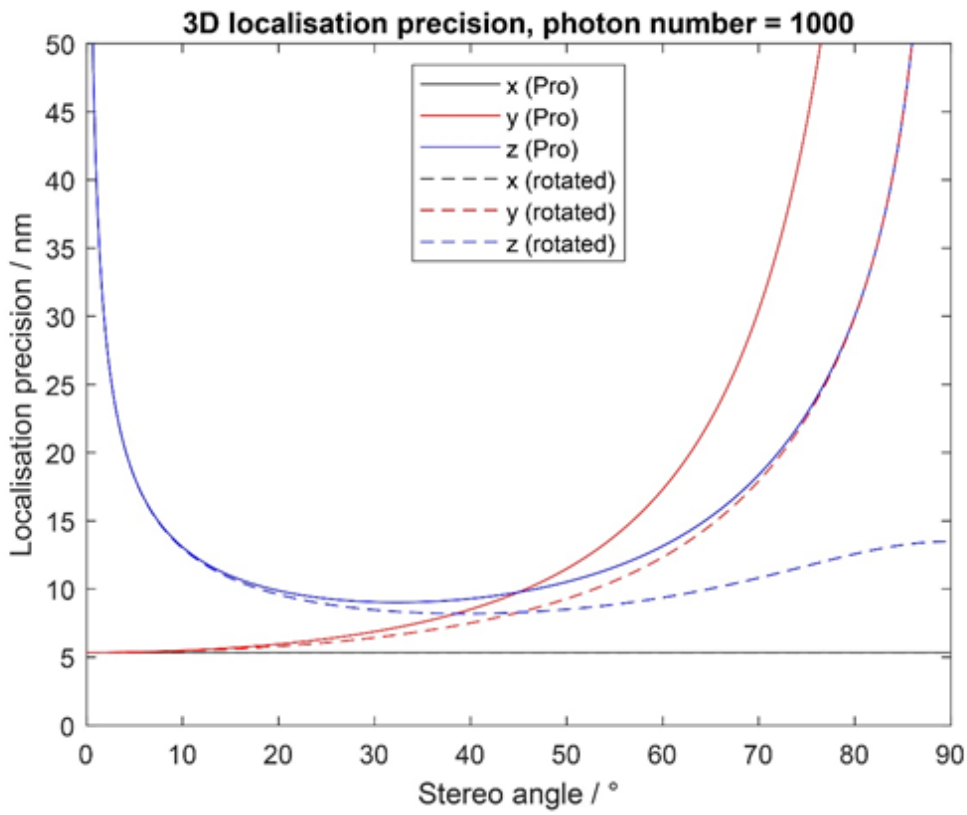
Here, the x value is estimated as before and so has the same variance, but now

$$
\begin{aligned}
\sigma_y^2 &= \frac{\left(\sigma_{y'}^{\mathrm{rot}}\right)^2}{2\cos\theta}, \\
\sigma_z^2 &= \frac{\left(\sigma_{y'}^{\mathrm{rot}}\right)^2}{2\sin\theta}.
\end{aligned}
$$

Supplementary Figure SN7.2 shows localization precisions for our projection method and rotated views over the angular range ±90°, given 500 photons per view. As expected, at small stereo angles the localization precisions are approximately equal for our projection method and rotated views, as at small angles $\cos\theta \sim 1$ and $\sin\theta \sim \tan\theta$. Intuitively, this can be understood as a result of the difference between our projection acquisition and rotated acquisition being the stretching by $1/\cos\theta$, which is ~ 1 for small angles. We would also expect both the y and z localisation precisions in the projection case to tend towards infinitely poor at ±90°, as here the image is stretched infinitely far, which is indeed observed. While the rotated case is always better than the projection case, the difference in precision over 'usable' angles (i.e. ±45°) is only of the order of a few nanometers. Finally, we note that the stereo angle at which equal localization precision in x and y is achieved is ±45° in both our projection and rotated cases, irrespective of photon number (N) or inherent PSF widths (s). However, a ±45° stereo view is not necessarily the best approach if the goal is to instead minimize the total localization precision

$$
\sigma = \sqrt{\sigma_x^2 + \sigma_y^2 + \sigma_z^2},
$$

which does depend on the inherent PSF widths.



Supplementary Figure SN7.2 Thompson–Mortensen localization precisions for two views obtained by our projection method or rotated view imaging. Each view is $\pm$ the stereo angle from the conventional deskewed view, calculated using the bead FWHM measurements described in the main text. As there is no change along the x direction in either method, the x localisation precision is equal in all situations. In both cases, the stereo angle at which equal y and z precision is achieved is ±45°.

Supplementary Note 8: particle localization using projected views.

Diffraction limited objects were detected on the original projection and projection of the sheared volume using previously published approach[16]. The vertical shift induced by the volume shearing was computed through the vertical association of detected object across projections through the resolution of a linear assignment problem expressed as:

$$\underset{\{a_{ij}\}}{argmin} \sum_{i\in\Omega, j\in\Omega'} c_{ij}a_{ij} \;\; s.t. \sum_{i\in\Omega} a_{ij} = 1 \; and \sum_{j\in\Omega'} a_{ij} = 1,$$

where $a_{ij} \in \{0,1\}$ denotes the assignment of the ith detection in the original projection to the jth measurement in the projection of the sheared volume, and c_{ij} is defined as the vertical distance as

$$c_{ij} = \begin{cases} abs(\boldsymbol{y}_i - \boldsymbol{y}'_j)), & \boldsymbol{x}_i < r_x \\ +\infty, & x \geq r_x \end{cases}$$

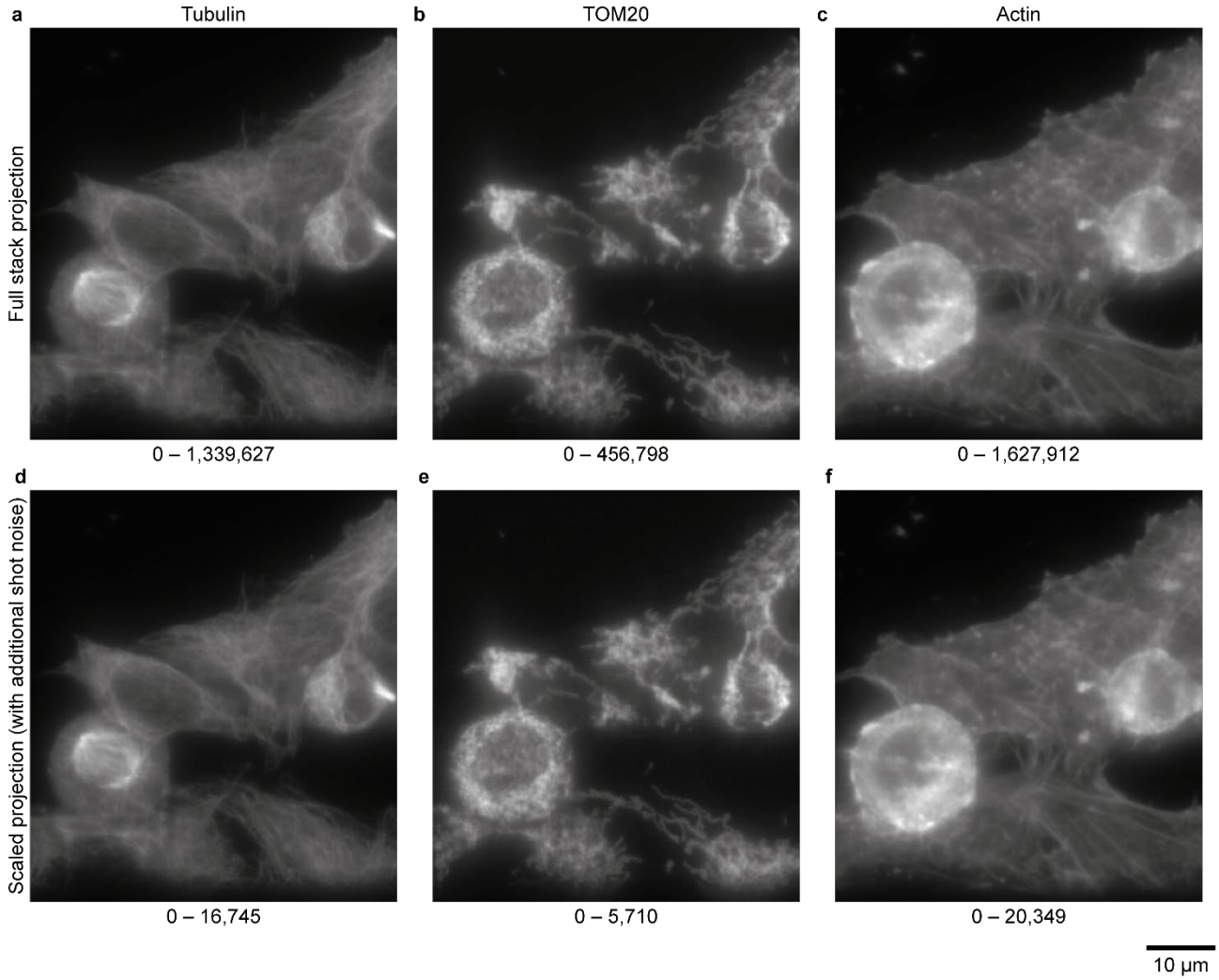
Where $\boldsymbol{y}'_j$ denotes the position in the projection of the sheared volume and r_x is a lateral gating parameter set to 3. For each optimal pair, the associated depth z_i is estimated as $z_i = c_{ij}/\tan(\alpha)$ where alpha is the shearing angle. The ground truth was computed using the same detection approaches implemented in 3D and aligned using a gradient based optimization. An estimated position is considered as a true positive if it is matched to an object in the ground truth with distance smaller than 575nm. The associated Jaccard Index (computed as TP/(|Ω|+|Ω'|-TP) where TP is the count of true positives) is 0.56.

Supplementary Note 9: dynamic ranges of optical and computational projections

Most scientific cameras used for microscopy feature a 16-bit dynamic range, with the sCMOS cameras used in this work having a gain of ~0.48 e^-/ADU. In normal 3D imaging, each pixel is cleared of photoelectrons after each slice has been imaged, while in projection imaging pixels are only cleared after the full volume has been illuminated. To investigate whether this may ever be a limiting factor in optical projection imaging, we obtained high-SNR 3D stacks of immunostained HeLa cells using a high-resolution light sheet microscope and computationally projected them. This produced maximum intensity values of 1339627, 456798 and 1627912 for the tubulin, TOM20 and actin channels, respectively. These values correspond to dynamic ranges of 20.35 bits, 18.8 bits and 20.6 bits, respectively and so it appears that a 16-bit dynamic range would be insufficient.

However, these stacks were acquired using a 50 ms per-slice exposure time, 396 slices and 5 mW of laser power (as measured in the pupil), corresponding to a 19.8 s acquisition time and 99 mJ of energy deposited into the sample. For live-cell imaging, typical pupil powers are ~50 µW (i.e. 100-fold lower), while acquisition times are of the order of a few seconds. Dividing the maximum intensity value measured by 20 (to simulate a conservatively lowered excitation power) and again by 4 (to simulate a ~5 s acquisition time) produces a maximum intensity of 20349, which comfortably fits within a 16-bit dynamic range (65336). Hence, it appears very unlikely that live-cell imaging in optical projection mode will ever saturate a 16-bit camera. Even if saturation does occur for high-power, long exposure fixed-cell imaging, this can always be compensated for by lower the excitation power or the exposure time. Fully saturating the sCMOS camera used would require ~31500 photons per pixel, corresponding to a SNR of better than 175.

Supplementary Figure SN9.1 shows the computationally projected stacks alongside versions which have had their photon counts reduced by a factor of 80 and appropriate shot noise added, to simulate the equivalent optical projection described above. Despite the additional shot noise, no visual distinction between the two can be made, due to the high SNR in both cases. Indeed, most computer displays are limited to 8 bits of dynamic range, resulting in a large compression of dynamic range for display purposes.



Supplementary Figure SN9.1 – Dynamic range comparison for full stack projection and optical projection. Top row: computational projections of 3D stacks of **a** tubulin, **b** TOM20, **c** actin, with full intensity ranges shown below. Bottom row: simulated optical projections given 80-fold less signal, as described in Supplementary Note 9, for **d** tubulin, **e** TOM20, **f** actin.

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Supplementary Tables

Figure	microscope	Exposure time	Acquisition rate	Voxel size	Laser power	Post processing
1D&G	Field Synthesis	25ms	NAN	104x104x300 nm^3	80µW	Raw data
1H	Field Synthesis	90ms	5Hz	104x104 nm^2	80µW	Raw data
2A	Field Synthesis	25ms	NAN	104x104x300 nm^2	80µW	2D deconvolution
2B	Field Synthesis	90ms	5Hz	104x104 nm^3	80µW	2D deconvolution, two frame average and Gaussian filter
2C	mSPIM	200ms	4Hz	162.5x162.5 nm^2	1.6mW	Brightness/Contrast correction
2D	mSPIM	200ms	4Hz	406x406 nm^2	0.3mW	Brightness/Contrast correction
2E	Spinning disk	200ms	4Hz	57x57 nm^2	350µW	Raw data
2F-G	OPM	100ms	10Hz	115x115 nm^2	410µW	Raw data
3A-E	Field Synthesis	90ms	5Hz	104x104 nm^2	80µW	2D deconvolution, five frame average and Gaussian filter (A-C) and raw data (D,E)
3F-G	OPM	40ms	12.5Hz	115x115 nm^2	750µW	Raw data
4B	Field Synthesis	250ms	1Hz	104x104 nm^2	50µW	anaglyph
4C	OPM	400ms	NA	115x115 nm^2	410µW	Raw data and 3D reconstruction
4F	Field Synthesis	100ms	10Hz	104x104 nm^2	30µW	Raw data
Video 1	Animation	NA	NA	NA	NA	NA
Video 2	Field Synthesis	100ms	10Hz	104x104 nm^2	80µW	Raw data
Video 3	Field Synthesis	90ms	5Hz	104x104 nm^2	80µW	2D deconvolution and Gaussian filter
Video 4	mSPIM	200ms	4Hz	162.5x162.5 nm^2	1.6mW	Brightness/Contrast correction
Video 5	mSPIM	200ms	4Hz	406x406 nm^2	0.3mW	Brightness/Contrast correction
Video 6	Spinning disk	200ms	4Hz	57x57 nm^2	350µW	Brightness/Contrast correction

Video 7	**Spinning Disk**	**750ms**	**0.66Hz**	**103x103 nm^2**	**460µW (470nm) 685µW (555 nm)**	**Raw data**
Video 8	**OPM**	**100ms**	**10Hz**	**115x115 nm^2**	**410µW**	**Raw data**
Video 9	**Field Synthesis**	**90ms**	**5Hz**	**104x104 nm^2**	**80µW**	**2D deconvolution and Gaussian filter**
Video 10	**OPM**	**40ms**	**12.5Hz**	**115x115 nm**	**750µW**	**2D deconvolution**
Video 11	**OPM**	**50ms**	**20Hz**	**115x115 nm^2**	**750µW**	**Gaussian filter (sigma 1 pixel)**
Video 12	**OPM**	**50ms**	**20Hz**	**115x115 nm^2**	**1.65mW**	**2D deconvolution and Gaussian filter**
Video 13	**Field Synthesis**	**250ms**	**1Hz**	**104x104 nm^2**	**50µW**	**Raw data**
Video 14	**Field Synthesis**	**100ms**	**10Hz**	**104x104 nm^2**	**30µW**	**Raw data**
Video 15	**mSPIM**	**200ms/ 40ms**	**4Hz/ 20Hz**	**406x406 nm^2/ 406x406 nm^2**	**2.5mW**	**Brightness/Contrast correction**
Video 16	**OPM**	**7.5ms**	**119Hz**	**115x115 nm^2**	**3.7mW**	**Gaussian filter (sigma 1 pixel)**

Supplementary Table 1 - Image acquisition parameter and post-processing of the image data. When a Gaussian filter was used, a width (sigma) of one pixel was applied.

Chipsize/Camera	**Orca flash 4.0**	**Kinetix**	**Orca fusion**	**Orca lightning**
1024X1024	80	260	200	76
512X1024	160	520	400	150
256X1024	320	1041	799	291

Supplementary Table 2-Framerates of current sCMOS cameras. Framerates (Hz) for different regions of interest for four contemporary sCMOS cameras.